

B. Sc. (Hons.) Semester - IV Examination 2023-24
STATISTICS

Paper No. STB-401: Sample Surveys and Design of Experiments

Time: Three Hours

Max. Marks: 70

NOTE: Attempt any five questions. All questions carry equal marks.

1. Write a brief note on the estimation of sample size in the case of simple random sampling addressing any two methods known to you.

For a population of size $N = 430$, we roughly know that $\bar{Y} = 19$ and $S^2 = 85.6$. With SRS, what should be the size of the sample to estimate $\bar{Y}$ with a margin of error of 10% of $\bar{Y}$, apart from a 1 in 20 chance?

2. What do you understand by error in sampling? Discuss, in detail, the different types of errors involved in sampling and how they can be controlled.
3. Explain the method of optimum allocation of sample sizes for different strata when the total sample size n is fixed.

A sampler proposes to take a stratified random sample. The field costs are expected to be of the form $\sum c_i n_i$ and the advance estimates of relevant quantities for two strata are as follows:

Stratum	$p_i = N_i/N$	S_i	c_i
1	0.4	10	4
2	0.6	20	9

Considering that the notations have their usual meanings:

- (a) Find the values of n_1/n and n_2/n that minimize the total cost for a given value of $V(\bar{y}_{st})$.
 - (b) Find the sample size required, under this optimum allocation, to make $V(\bar{y}_{st}) = 1$, if fpc is ignored.
 - (c) Obtain the total minimized cost.
4. Discuss the concept of systematic sampling with its merits, demerits, and real-life uses. Obtain the algebraic expression for the variance of the estimated mean under systematic sampling.
 5. What do you mean by the regression method of estimation? Find out the bias and variance of the linear regression estimator for the population mean. Also prove that the regression estimator for the population mean is always more efficient than or equal to the ratio estimator.
 6. What do you understand by Analysis of Variance (ANOVA)? Mention the basic assumptions underlying the ANOVA F -test. Describe the ANOVA technique for a fixed-effect model for two-way classified data with one observation per cell.
 7. Explain the fundamental principles of design of experiments (Randomization, Replication, and Local Control) with clear illustrations. Also discuss the layout of a Randomized Block Design (RBD) along with its structural disadvantages.
 8. Write a comprehensive note on the design which follows all three basic principles of design of experiments (Latin Square Design). Mention its operational advantages and disadvantages over other basic designs.
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- **StudentResourceCenter:** <https://quashan-me.vercel.app/>
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